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# FIND-S Algorithm in Google Colab

# Training Data

# Attributes: Sky, AirTemp, Humidity, Wind, Water, Forecast, EnjoySport

training_data = [

    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],

    ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes'],

    ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No'],

    ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']

]

# Display Training Data

print("Training Data:\n")

for row in training_data:

    print(row)

# FIND-S Algorithm

hypothesis = ['0'] * (len(training_data[0]) - 1)

print("\nSteps of FIND-S Algorithm:\n")

for i, sample in enumerate(training_data):

    if sample[-1] == "Yes":

        for j in range(len(hypothesis)):

            if hypothesis[j] == '0':

                hypothesis[j] = sample[j]

            elif hypothesis[j] != sample[j]:

                hypothesis[j] = '?'

        print(f"Step {i+1}: {hypothesis}")

print("\nFinal Most Specific Hypothesis:")

print(hypothesis)

```

Training Data:

```

***
[['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes']
 ['Sunny', 'Warm', 'High', 'Strong', 'Warm', 'Same', 'Yes']
 ['Rainy', 'Cold', 'High', 'Strong', 'Warm', 'Change', 'No']
 ['Sunny', 'Warm', 'High', 'Strong', 'Cool', 'Change', 'Yes']]

```

Steps of FIND-S Algorithm:

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Step 1: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
Step 2: ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
Step 4: ['Sunny', 'Warm', '?', 'Strong', '?', '?']

```

Final Most Specific Hypothesis:

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['Sunny', 'Warm', '?', 'Strong', '?', '?']

```